

Combining On-Policy Optimization and Distillation for Long-Context Reasoning in Large Language Models

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Abstract

Adapting large language models (LLMs) to long-context tasks requires post-training methods that remain accurate and coherent over thousands of tokens. Existing approaches are limited in several ways: 1) off-policy methods such as supervised fine-tuning (SFT) and knowledge distillation (KD) suffer from exposure bias and limited recovery from model-generated errors over long horizons; 2) on-policy reinforcement learning methods such as Group Relative Policy Optimization (GRPO) better align training with model-generated states, but are unstable and sample-inefficient due to sparse rewards; 3) on-policy distillation (OPD) provides dense token-level guidance, but does not directly optimize arbitrary reward signals. In this paper, we propose Distilled Group Relative Policy Optimization (dGRPO), a method for long-context reasoning that augments GRPO with dense guidance from a stronger teacher via OPD. We also introduce **LongBlocks**, a synthetic long-context dataset spanning multi-hop reasoning, contextual grounding, and long-form generation. We conduct extensive experiments and ablations comparing off-policy training, sparse-reward GRPO, and our combined approach, leading to an improved recipe for long-context alignment. Overall, our results show that combining outcome-based policy optimization with knowledge distillation in a single objective provides a more stable and effective path to long-context reasoning, while preserving short-context capabilities.¹

1 Introduction

Real-world applications increasingly require large language models (LLMs) to reason over and integrate information from extremely long contexts, such as understanding large codebases (Jimenez et al., 2024), synthesizing information across collections of documents (Yang et al., 2018), and maintaining coherence across multi-session interactions spanning hundreds of thousands of tokens (Mei et al., 2025; Liu et al., 2024). Yet, aligning LLMs for long-context reasoning remains challenging due to multiple factors, including positional extrapolation, attention dilution, and exposure bias (Peng et al., 2024; Liu et al., 2024; Bengio et al., 2015). Existing post-training strategies lack a standardized, open recipe that reliably delivers strong long-context performance (OpenAI, 2025; Yang et al., 2025; Comanici et al., 2025). Current efforts are further constrained by limited data and compute: off-policy alignment datasets are typically small and focused on relatively simple retrieval and summarization tasks, which restricts the diversity of long-context supervision (Bai et al., 2024a; Chen et al., 2024). At the same time, long-context settings remain relatively underexplored for on-policy alignment (Ouyang et al., 2022) and distillation (Agarwal et al., 2024). Existing long-context work has largely focused on preference optimization (Bai et al., 2024c; Zhang et al., 2025) or on-policy methods with limited sampling (Wan et al., 2025), leaving the use of richer reward signals poorly understood in long-context regimes.

In this work, we ask the following question: **which post-training ingredients most effectively enable efficient, stable, and generalizable long-context reasoning?** We address

¹All resources are available on [Hugging Face](#).

Method	Trajectory Sampling	Learning Signal	Reward Objective	Exposure Bias
SFT / KD (Hinton et al., 2015)	✗ Off-policy	✓ Dense	✗ No	✗ High
OPD (Agarwal et al., 2024)	✓ On-policy	✓ Dense	✗ No	✓ Reduced
GRPO (Shao et al., 2024)	✓ On-policy	✗ Sparse	✓ Yes	✓ Low
dGRPO (ours)	✓ On-policy	✓ Dense	✓ Yes	✓ Low

Table 1: Comparison of post-training strategies considered in this work.

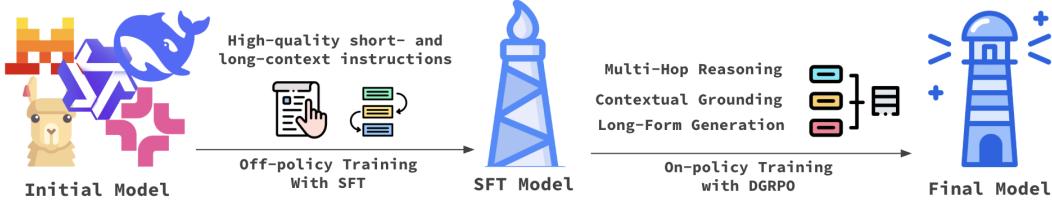


Figure 1: Illustration of our post-training recipe for building a long-context reasoning model.

the limitations of existing methods by proposing **Distilled Group Relative Policy Optimization (dGRPO)**, which combines the objectives of GRPO and on-policy distillation (OPD) to provide both reward-driven optimization and dense teacher guidance during on-policy learning. While inspired by prior work on on-policy distillation (Agarwal et al., 2024), this approach remains insufficiently studied for long-context reasoning tasks. Table 1 summarizes the properties of the post-training strategies considered in this work.

To evaluate this approach, we conduct a rigorous comparative empirical study of long-context alignment strategies in a Qwen3-based setup (Figure 1). We derive from these findings a practical post-training recipe for long-context reasoning, and introduce **LONG-BLOCKS**, a synthetically generated dataset for long-context training comprising multi-hop reasoning, contextual grounding, and long-form generation examples, to support this recipe. Together, these components yield consistent improvements in long-context performance while preserving the existing short-context capabilities of the starting model.

2 Preliminary

2.1 Token-level MDP for Language Generation

We formulate autoregressive language generation as a token-level MDP $\mathcal{M} = (S, A, r, T)$. At each generation step t , the state $s_t \in S$ consists of the prompt p concatenated with all previously generated tokens: $s_t = [p_1, \dots, p_M, o_1, \dots, o_{t-1}]$. The action at time t is the next token $o_t \in A$ selected from the model vocabulary $\mathcal{V}$. Transitions are deterministic, with each new state formed by appending the new token: $s_{t+1} = [s_t; o_t]$. The initial state s_1 is induced by the prompt $p \sim \mathcal{P}$, where $\mathcal{P}$ denotes the distribution over prompts. An episode ends when the policy outputs an end-of-sequence token [eos] or when the total token budget is reached. A policy is an autoregressive conditional distribution $\pi_\theta(\cdot|s_t)$, and the reward for a trajectory $\tau = (p, o)$ is the sum of per-step rewards: $R(p, o) = \sum_{t=1}^{|o|} r(s_t, o_t)$. We consider the undiscounted finite-horizon setting, which is standard for sequence generation tasks. The agent typically maximizes a KL-regularized expected return (Schulman et al., 2018):

$$\mathcal{J}(\pi_\theta) = \mathbb{E}_{p \sim \mathcal{P}, o \sim \pi_\theta(\cdot|p)} \left[R(p, o) - \beta_{\text{ref}} \sum_{t=1}^{|o|} D_{\text{KL}}(\pi_\theta(\cdot|s_t) \parallel \pi_{\text{ref}}(\cdot|s_t)) \right]. \quad (1)$$

Here π_{ref} is a reference policy, and β_{ref} is a KL regularization weight used in RLHF to prevent the tuned policy from drifting into regions where the learned reward model is unreliable. Following prior work (Liu et al., 2025; Yu et al., 2025; Olmo et al., 2025; Team et al., 2025), we set $\beta_{\text{ref}} = 0$, eliminating the need for a reference policy and reducing memory and compute costs. The KL penalty term is typically used to prevent the online policy from diverging too far from the reference policy. However, for reasoning models, the model distribution can naturally diverge from the initial model without harming performance.

2.2 Off-Policy Training

Off-policy supervision, primarily via supervised fine-tuning (SFT), provides a stable and data-efficient approach to bootstrap long-context behavior and can be viewed as a form of imitation learning (Ross et al., 2011) for language models. Formally, given an expert dataset $\mathcal{D}$ of input–output pairs (x, y) , we optimize the model parameters θ by minimizing the token-level negative log-likelihood:

$$\mathcal{L}_{\text{SFT}}(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[- \sum_{t=1}^{|y|} \log \pi_{\theta}(y_t \mid x, y_{<t}) \right]. \quad (2)$$

This objective delivers dense, token-level supervision, making it computationally efficient and effective for instilling task adherence and structural coherence (Raffel et al., 2020; Ouyang et al., 2022). However, when a strong teacher model is available, knowledge distillation (Hinton et al., 2015) can further enhance learning by aligning the student policy π_{θ} with the teacher policy π_{teacher} . This is typically achieved by minimizing the token-level Kullback–Leibler divergence between the teacher and student distributions:

$$\mathcal{L}_{\text{KD}}(\theta) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\sum_{t=1}^{|y|} D_{\text{KL}}(\pi_{\text{teacher}}(\cdot \mid x, y_{<t}) \parallel \pi_{\theta}(\cdot \mid x, y_{<t})) \right]. \quad (3)$$

This distillation objective encourages the student to match the teacher’s token distributions at each prefix, providing fine-grained guidance that can improve generalization and long-range reasoning beyond standard SFT. However, relying on static expert trajectories limits both off-policy methods due to *exposure bias* (Ranzato et al., 2016; Bengio et al., 2015)—they optimize only on prefixes drawn from the data distribution and therefore do not directly train recovery from the model’s own off-distribution prefixes. Multiple studies show that exposure bias can degrade generation quality by amplifying errors during inference (Arora et al., 2022; Zhang et al., 2019; Chiang & Chen, 2021). In long-context settings, this issue can interact with other sources of degradation, including distractor-heavy retrieval, positional extrapolation, answer-length calibration, and optimization instability at long sequence lengths. As a result, without mechanisms to correct for off-distribution states, these methods can produce unstable or suboptimal generations in long-context tasks.

2.3 On-Policy Training

On-policy RL approaches, such as *Group Relative Policy Optimization* (GRPO) (Shao et al., 2024), address an important limitation of off-policy training approaches by conditioning updates on the model’s own rollouts. GRPO avoids exposure bias by optimizing the policy under the distribution induced by its own rollouts, enabling adaptation to long-range dependencies. Unlike standard PPO (Schulman et al., 2017), GRPO simplifies value estimation by removing the need for a separate value function critic. Instead, it estimates the baseline from a group of sampled completions $\{o_1, o_2, \dots, o_G\}$ for a given prompt p . The objective function, with $\beta_{\text{ref}} = 0$, can be defined as:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{p \sim \mathcal{P}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot \mid p)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(\rho_{i,t}(\theta) \hat{A}_{i,t}, \bar{\rho}_{i,t}(\theta) \hat{A}_{i,t} \right) \right], \quad (4)$$

where

$$\rho_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t} \mid p, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid p, o_{i,<t})}, \quad \bar{\rho}_{i,t}(\theta) = \text{clip}(\rho_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon),$$

and

$$\hat{A}_{i,t} = \frac{R(p, o_i) - \text{mean}(\{R(p, o_1), \dots, R(p, o_G)\})}{\text{std}(\{R(p, o_1), \dots, R(p, o_G)\})}.$$

Nevertheless, standalone GRPO can be sample-inefficient and high-variance due to sparse reward signals, and remains susceptible to reward hacking (Liu et al., 2025; Yu et al., 2025), especially during long-context training where trajectory-level rewards are delayed and the

lack of fine-grained supervision can hurt stability. An alternative approach that provides denser supervision is *on-policy distillation* (OPD), which guides the student policy toward a strong teacher by minimizing the reverse token-level KL divergence on student-generated trajectories, thereby providing token-level guidance while remaining on-policy (Agarwal et al., 2024; Yang et al., 2025; Lu & Lab, 2025). Unlike regularization toward a fixed reference policy, OPD aligns the student to the teacher on the prefixes induced by the student’s own rollouts. Using reverse KL makes the distillation objective student-centered, focusing supervision on the trajectories the student actually visits, which makes it a natural fit for on-policy RL. Formally, we define the on-policy distillation objective as:

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{p \sim \mathcal{P}, o \sim \pi_{\theta}(\cdot | p)} \left[\frac{1}{|o|} \sum_{t=1}^{|o|} D_{\text{KL}}(\pi_{\theta}(\cdot | p, o_{<t}) \parallel \pi_{\text{teacher}}(\cdot | p, o_{<t})) \right]. \quad (5)$$

However, OPD alone does not perform reward optimization and, therefore, cannot fully exploit reward-based feedback. In Section 3.1, we address this limitation with dGRPO, which combines on-policy reward optimization and dense distillation.

3 Recipe for Long-Context Reasoning Alignment

3.1 Training Pipeline for Improving and Extending Long-Context Capabilities

As illustrated in Figure 1, we use a unified training pipeline to scale RL to long-context reasoning. Unlike standard RLHF pipelines, which often struggle at extended context lengths due to sparse rewards, unstable on-policy rollouts, and degradation of reasoning structure, this pipeline combines long-horizon reward optimization with dense teacher guidance via dGRPO. This design enables stable policy improvement at extended sequence lengths while preserving short-context performance.

Cold-start Stage: Off-policy warm-up for stable long-horizon rollouts. The pipeline begins with an SFT warm-up stage that provides dense token-level supervision, improves the coherence of long-context rollouts, and stabilizes subsequent on-policy learning. This initialization is especially beneficial in long-horizon settings, where token-level advantage estimation and KL-regularized updates can become brittle when rollout quality is low or highly variable (Stiennon et al., 2020; Ouyang et al., 2022; Bai et al., 2022). We demonstrate the benefits of this cold-start stage empirically in the ablation experiments described in §4.3 and Appendix C.5, including comparisons between SFT- and KD-based initialization.

RL Stage: On-policy optimization with Distilled Group Relative Policy Optimization (dGRPO). After initialization, we perform on-policy RL with GRPO (Eq. 4) to optimize long-horizon task rewards under the model’s own rollout distribution. In long-context settings, however, optimizing sparse rewards alone can degrade reasoning structure and lead to forgetting of short-context skills. To mitigate these effects, we propose Distilled Group Relative Policy Optimization (dGRPO), a variant of GRPO with teacher regularization inspired by prior work by Agarwal et al. (2024) showing that RLHF can be combined with on-policy distillation. Concretely, dGRPO combines two complementary signals: sparse outcome-level rewards from GRPO and dense token-level guidance from the teacher. Given a prompt $p \sim \mathcal{P}$, we sample a group of G rollouts $\{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot | p)$ and compute normalized advantages $\hat{A}_{i,t}$. Using the importance ratios $\rho_{i,t}(\theta)$ and clipped ratios $\bar{\rho}_{i,t}(\theta)$ from Eq. 4, we maximize

$$\begin{aligned} \mathcal{J}_{\text{dGRPO}}(\theta) = \mathbb{E}_{p \sim \mathcal{P}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot | p)} & \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min(\rho_{i,t}(\theta) \hat{A}_{i,t}, \bar{\rho}_{i,t}(\theta) \hat{A}_{i,t}) \right. \\ & \left. - \beta D_{\text{KL}}(\pi_{\theta}(\cdot | p, o_{i,<t}) \parallel \pi_{\text{teacher}}(\cdot | p, o_{i,<t})) \right], \end{aligned} \quad (6)$$

where $\beta \geq 0$ controls the strength of teacher regularization. The first term is the GRPO objective (Eq. 4) for optimizing sparse rewards, and the KL term from OPD (Eq. 5) provides token-level feedback, anchoring the student to the teacher along its own sampled trajectories.

Source	Coverage	Samples
Institutional Books	3 natural languages (de, en, fr)	107,908
FineWeb2-HQ	9 natural languages (de, el, es, fr, hu, it, nl, sv, zh)	39,113
ArXiv	English	13,969
Project Gutenberg Books	10 natural languages (de, en, fr, hu, it, nl, pl, pt, sv, zh)	11,009
Wikipedia	30 natural languages (ar, ca, da, de, el, es, ..., uk, zh)	10,718
Stack-Edu	15 programming languages (Python, C++, ..., Markdown)	5,430
StackExchange	English	5,072
Total		193,219

Table 2: Sources used to generate multilingual synthetic long-context data for **LongBlocks**. Appendix C.3 details the language-wise composition of **LongBlocks**.

3.2 **LongBlocks**: A Multilingual Dataset to Tailor LLMs for Long-Context Reasoning

To support long-context training in LLMs, we introduce **LongBlocks**, a multilingual synthetic dataset for long-context adaptation. In this work, we use it as a training resource for both off-policy warm-up and on-policy alignment. **LongBlocks** contains 193,219 question-answer pairs generated from raw documents spanning diverse sources, including ArXiv (Clement et al., 2019), Wikipedia (Foundation), StackExchange (Kandpal et al., 2025), Project Gutenberg Books (Gutenberg), Stack-Edu (Allal et al., 2025), FineWeb2-HQ (Messmer et al., 2025), and Institutional-Books-1.0 (Cargnelutti et al., 2025). We construct **LongBlocks** with a two-stage pipeline. First, we sample sufficiently long, self-contained documents from these corpora and prompt **Qwen3-Next-80B-A3B-Thinking** with the templates in Appendix A to generate candidate question-answer pairs that require integrating evidence across extended contexts. Second, we remove malformed and near-duplicate documents using fuzzy deduplication and verify candidate synthetic examples with an LLM-as-a-judge step, again using **Qwen3-Next-80B-A3B-Thinking**. This verification rejects malformed or weakly grounded examples, including incomplete or unsupported answers and questions answerable from a local span. **LongBlocks** covers more than 30 natural languages and 15 programming languages. Table 2 summarizes the sources, languages, and sample counts.

To mitigate catastrophic forgetting across context lengths, we maintain a fixed 10/90 token-level mixture of short- and long-context data during both SFT and RL. Short-context samples are drawn uniformly from **Nemotron-Post-Training-Dataset-v2**, whereas long-context samples are drawn from **LongBlocks**. We analyze the effect of this mixture in Appendix C.4.

3.3 Training Details

We initialize from **Qwen3-1.7B** and apply the two-stage post-training pipeline described in §3.1: SFT followed by RL. We extend the model to 128K context without modifying the original positional parameterization and fix the RoPE base frequency to 1×10^6 (Su et al., 2024). For SFT, we train for two epochs on the data mixture described above, using a learning rate of 1×10^{-5} and a batch size of four million tokens. Optimization uses AdamW (Kingma & Ba, 2015) with weight decay 0.01. We use bfloat16 mixed precision, sequence packing (Raffel et al., 2020), and document masking to prevent cross-document attention (Grattafiori et al., 2024). Cross-entropy loss is computed only on target tokens. For RL, we run dGRPO for one epoch on the same data mixture. Following Olmo et al. (2025), we use verifiable rewards for short-context tasks, including SymPy verification for math, unit tests for coding, and binary constraint satisfaction for instruction-following tasks. For chat and long-context tasks, we follow Lee et al. (2024) and use **Qwen3-4B** as a reference-conditioned LM judge that assigns binary rewards by comparing responses against ground-truth answers rather than relying on open-ended preference judgments (Appendix B). We use **Qwen3-32B** as the teacher model and set $\beta = 0.5$. We analyze the sensitivity to β in §4.3. Optimization uses Adam with learning rate 1×10^{-6} , $(\beta_1, \beta_2) = (0.9, 0.95)$, gradient clipping of 1.0, a constant learning-rate schedule, a clipping threshold of 0.2, and 8 sampled generations per prompt.

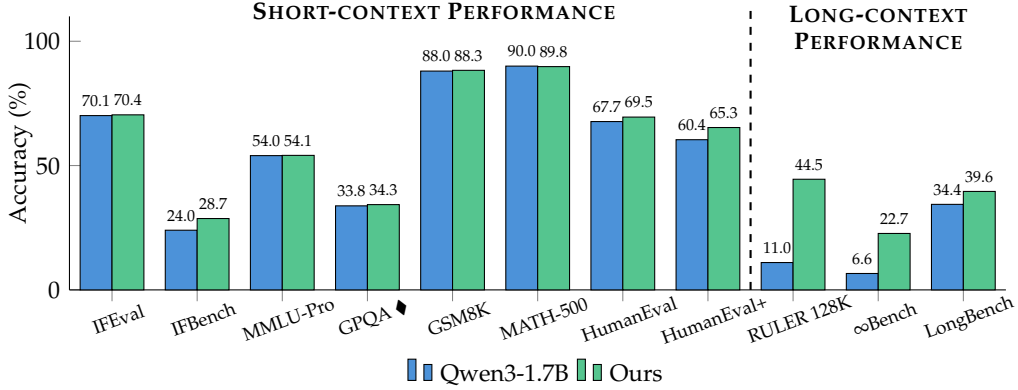


Figure 2: Short- and long-context benchmark suite.

4 Experiments

4.1 Experimental Setup

We compare our final model against the base model, Qwen3-1.7B, an SFT-only checkpoint from the same training pipeline, and strong external long-context baselines.

Decoding configuration. All models are evaluated with `lm-eval-harness` using a common decoding configuration: temperature 0.6, top- $p = 0.95$, top- $k = 20$, presence penalty 1.5, and a maximum generation length of 32,768 tokens, following Yang et al. (2025). To reduce variance from stochastic decoding, we run each benchmark five times and report the mean, with standard deviations across runs provided in Appendix C.6, as in Olmo et al. (2025).

Short-context evaluation. We evaluate short-context performance on IFEval (Zhou et al., 2023), IFBench (Pyatkin et al., 2025), MMLU-Pro (Wang et al., 2024), GPQA ♦ (Rein et al., 2024), GSM8K (Cobbe et al., 2021), MATH-500 (Lightman et al., 2024), HumanEval (Chen et al., 2021), HumanEval+ (Liu et al., 2023), and the multilingual benchmarks MMMLU (OpenAI, 2024) and MGSM (Shi et al., 2022). Together, these benchmarks assess instruction following, multi-step reasoning, multilingual capabilities, and code-generation correctness.

Long-context evaluation. We evaluate long-context performance at a context length of 128K tokens on RULER (Hsieh et al., 2024), OneRuler (Kim et al., 2025), LongBench (Bai et al., 2024b), and ∞Bench (Zhang et al., 2024). These benchmarks assess retrieval, diagnostic long-context tasks, multilingual aggregation, and end-to-end long-context performance.

4.2 Main Results and Findings

Our recipe achieves robust scaling to long-context tasks without degradation in short-context tasks. As shown in Figure 2 and Table 3, our recipe yields substantial gains on long-context tasks while preserving, and in some cases improving, short-context performance. Across short-context benchmarks, accuracy remains stable or improves slightly, with the largest gains appearing in code generation, suggesting that long-context alignment preserves capabilities and may strengthen reasoning. On long-context benchmarks, improvements are consistent: average RULER performance rises from 64.5 to 74.8, with gains increasing as sequence length grows and becoming especially pronounced at 64K and 128K tokens (11.0→44.5 at 128K), demonstrating effective scaling of retrieval and reasoning with context size. Similar trends hold on ∞Bench and LongBench, indicating that these improvements generalize across benchmarks. Appendix C provides detailed per-task and multilingual results, further confirming preservation of both short- and long-context capabilities. It also reports the ablation on balancing short- and long-context data (Section C.4), the cold-start SFT-vs.-KD ablation (Section C.5), and uncertainty across evaluation runs (Section C.6).

Model	RULER						
	Avg.	4K	8K	16K	32K	64K	128K
Qwen3-1.7B	64.5	87.4	84.2	81.0	72.0	51.5	11.0
Ours	74.8	88.3	85.4	84.7	80.4	65.6	44.5

Table 3: RULER performance aggregated across increasing sequence lengths.

Our recipe shifts the short-/long-context performance frontier. Figure 3 shows that our recipe shifts the short-/long-context performance frontier beyond both the base model and stronger external long-context baselines. This suggests genuine alignment gains rather than a trade-off of short-context performance for long-context ability. The figure also isolates each training stage’s effect. The SFT-only checkpoint improves long-context performance while preserving short-context competence, and the subsequent dGRPO stage adds substantial further gains. By contrast, applying dGRPO directly to the base model (RL-Zero), without the SFT warm start, produces a worse short-/long-context trade-off, underscoring Stage 1’s role in stabilizing long-horizon on-policy optimization. The final model is competitive with larger 3B–4B models, including a YaRN-based context-extension model from the same family, indicating that the gains stem from our recipe rather than scale.

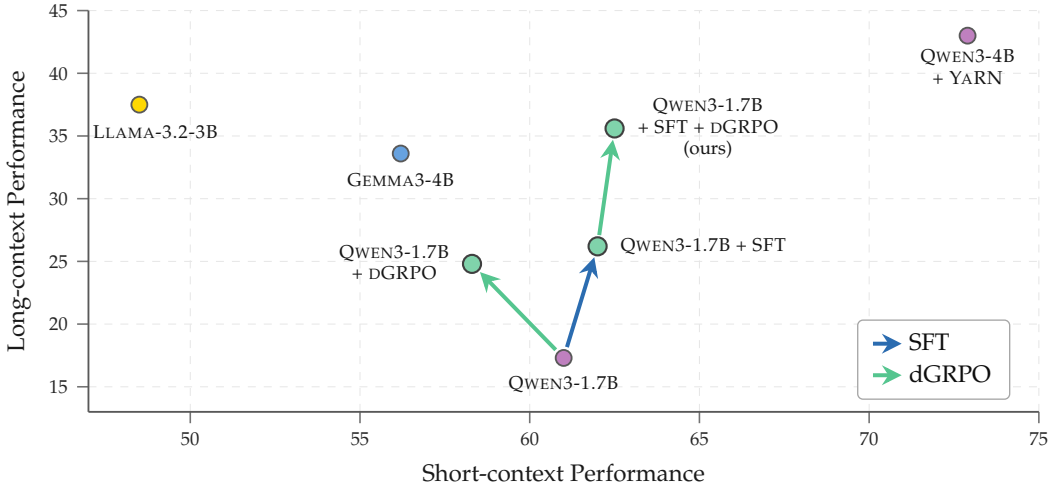


Figure 3: Short-/long-context frontier, averaging all benchmarks per context regime.

4.3 Dissecting the Training Recipe

DGRPO stabilizes on-policy optimization and improves long-context reasoning. Figure 4 compares GRPO, OPD, and dGRPO under the same SFT warm start and RL budget. GRPO increases reward but is much noisier, underscoring the challenge of optimizing long-context behavior from sparse trajectory-level rewards. OPD offers denser token-level supervision and smoother updates, but yields smaller gains on long-context tasks. By combining reward optimization with on-policy teacher regularization, dGRPO achieves the smoothest training, the highest final reward, better long-context performance, and nearly unchanged short-context accuracy. Together, these results provide additional evidence that OPD is an effective regularizer for on-policy training in long-context settings.

Effect of the teacher model in dGRPO: self-distillation vs. standard distillation. We study how the choice of teacher affects dGRPO by comparing distillation from a stronger external teacher with self-distillation from the current policy (Zhao et al., 2026), thereby isolating the impact of teacher-signal quality in dGRPO. While self-distillation already outperforms the baseline, using a stronger external teacher leads to substantially larger improvements, showing that dGRPO benefits directly from higher-quality teacher guidance.

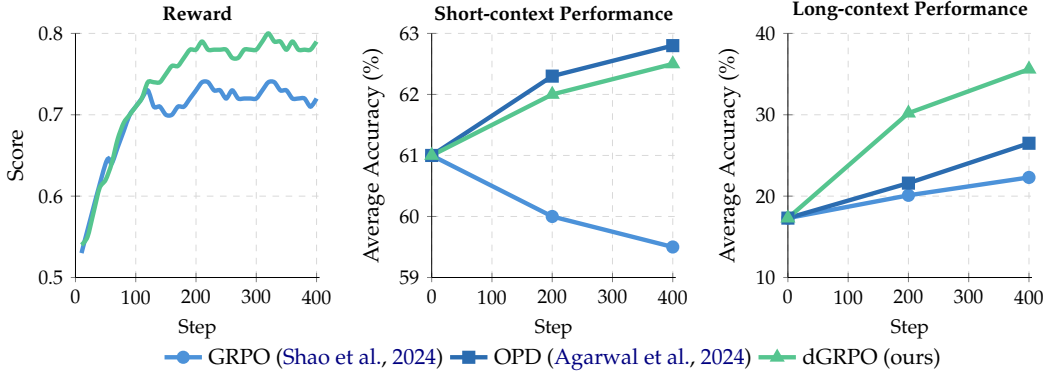


Figure 4: Comparison of GRPO, OPD, and dGRPO in terms of reward trajectories and evaluation performance across short- and long-context benchmarks.

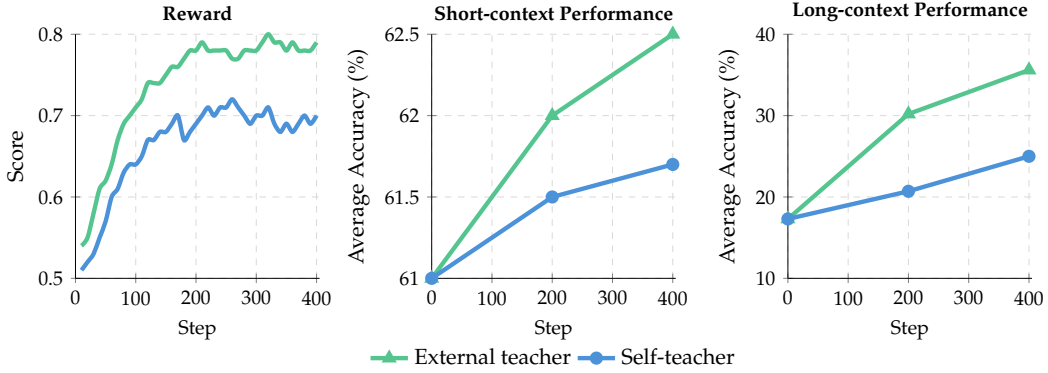


Figure 5: Comparison of external-teacher and self-teacher variants of dGRPO in terms of reward trajectories and evaluation performance across short- and long-context benchmarks.

Effect of β in dGRPO: balancing regularization and exploration. Figure 6 shows how the regularization weight β affects dGRPO (Eq. 6). At $\beta = 0$, dGRPO reduces to GRPO and performs worst, highlighting the difficulty of sparse-reward optimization without teacher regularization. Increasing β strengthens on-policy distillation, providing dense token-level guidance along model-generated trajectories and improving both short- and long-context performance. Performance peaks at $\beta = 0.5$, indicating the best balance between regularization and exploration in our long-context setting. This is consistent with Agarwal et al. (2024), who note that the optimal distillation strength is task-dependent.

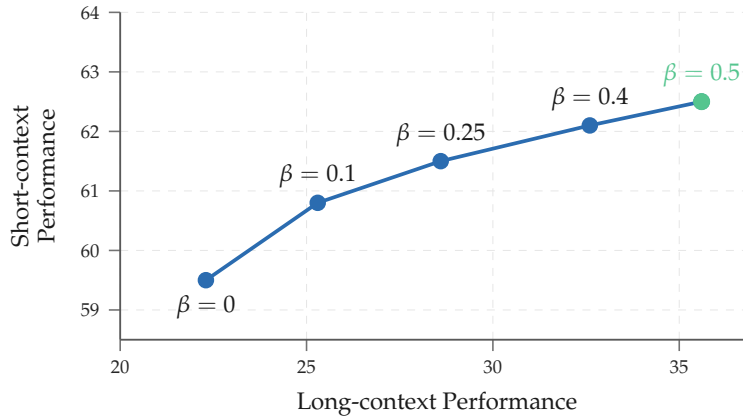


Figure 6: Effect of distillation strength β in dGRPO on short-/long-context performance.

5 Related Work

Alignment Methods: Off-Policy vs. On-Policy. Alignment methods for language models broadly fall into two categories. Off-policy methods, including supervised fine-tuning, distillation, and offline preference optimization such as DPO (Rafailov et al., 2024) and KTO (Ethayarajh et al., 2024), optimize on fixed datasets of demonstrations or preferences. They are typically stable and compute-efficient, but because they rely on static expert data rather than the model’s own rollouts, they can suffer from train–inference mismatch and exposure bias (Bengio et al., 2015; Ranzato et al., 2016). By contrast, on-policy methods optimize on samples generated by the current policy. This family includes reinforcement learning approaches such as PPO (Schulman et al., 2017), GRPO (Shao et al., 2024), DAPO (Yu et al., 2025), and Dr. GRPO (Liu et al., 2025), as well as on-policy distillation approaches (Agarwal et al., 2024; Yang et al., 2025; Zhao et al., 2026; Hübottter et al., 2026). This distinction matters especially for long-horizon generation, where local errors compound over extended rollouts.

Our recipe combines these complementary regimes through an off-policy cold start followed by on-policy optimization with dense teacher guidance.

Long-Context Adaptation and Post-Training. Prior work on long-context LLMs has largely focused on extending context windows and adapting models to long inputs. One line of work improves long-context capacity through architectural or positional modifications (Su et al., 2024; Chen et al., 2023; Peng et al., 2024). Another line of work adapts models to long inputs through off-policy post-training on static long-context data (Bai et al., 2024a; Chen et al., 2024). While effective, these approaches inherit the limitations of static supervision: they do not directly train the model on its own long-horizon failure modes and are often centered on relatively narrow task formats such as retrieval or summarization, even though strong long-context performance also requires robustness to distant dependencies and stable reasoning over long horizons (Liu et al., 2024; Bai et al., 2024b; Hsieh et al., 2024).

Our approach addresses this limitation by pairing long-context adaptation with broader supervision through LONGBLOCKS and optimization on model-generated trajectories.

On-Policy Long-Context Alignment. More recent work has begun to explore on-policy alignment for long-form generation and reasoning (Bai et al., 2024c; Zhang et al., 2025; Wan et al., 2025), showing that long-context post-training can benefit from reward-driven or preference-based optimization. However, these recipes remain narrowly scoped or tightly coupled to specific task families. In parallel, on-policy distillation has emerged as a promising way to provide dense token-level guidance on model-generated trajectories (Agarwal et al., 2024; Hübottter et al., 2026; Yang et al., 2025). Combining reward optimization with on-policy distillation, however, remains largely unexplored in long-context settings.

Our approach targets this gap with a recipe that culminates in dense teacher-guided on-policy optimization with dGRPO, yielding a more general long-context post-training recipe.

6 Conclusion

Long-context reasoning requires post-training methods that remain stable over long generation horizons while still adapting to the model’s own errors and distribution shifts at inference time. Our study highlights a practical tradeoff: off-policy methods are stable but limited by static supervision and exposure bias, whereas pure on-policy RL is better suited to long-horizon adaptation but becomes brittle under sparse trajectory-level rewards.

To address this challenge, we propose a two-stage recipe for long-context post-training: an off-policy cold start to improve long-context rollouts, followed by dense teacher-guided on-policy optimization with dGRPO. By combining sparse task rewards with dense token-level teacher supervision on model-generated trajectories, dGRPO enables more effective long-context optimization. We complement this recipe with LONGBLOCKS, a synthetic long-context training dataset used for both off-policy warm-up and on-policy alignment. Across our evaluation suite, this recipe consistently improves long-context performance without sacrificing short-context capabilities.

Acknowledgments

We thank the members of the SARDINE lab for their useful and constructive comments. This work was supported by the project DECOLLAGE (ERC-2022-CoG 101088763), by the Portuguese Recovery and Resilience Plan through project C645008882-00000055 (Center for Responsible AI), and by FCT/MECI through national funds and when applicable co-funded EU funds under UID/50008: Instituto de Telecomunicações.

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Alberto Magni, Paweł Stradomski, Tal Marian, Vlado Galić, Wenhui Chen, Hisham Husain, Achintya Singhal, Dominik Grewe, François-Xavier Aubet, Shuang Song, Lorenzo Blanco, Leland Rechis, Lewis Ho, Rich Munoz, Kelvin Zheng, Jessica Hamrick, Kevin Mather, Hagai Taitelbaum, Eliza Rutherford, Yun Lei, Kuangyuan Chen, Anand Shukla, Erica Moreira, Eric Doi, Berivan Isik, Nir Shabat, Dominika Rogozińska, Kashyap Kolipaka, Jason Chang, Eugen Vušak, Srinivasan Venkatachary, Shadi Noghabi, Tarun Bharti, Younghoon Jun, Aleksandr Zaks, Simon Green, Jeshwanth Challagundla, William Wong, Muqthar Mohammad, Dean Hirsch, Yong Cheng, Iftekhhar Naim, Lev Proleev, Damien Vincent, Aayush Singh, Maxim Krikun, Dilip Krishnan, Zoubin Ghahramani, Aviel Atias, Rajeev Aggarwal, Christo Kirov, Dimitrios Vytiniotis, Christy Koh, Alexandra Chronopoulou, Pawan Dogra, Vlad-Doru Ion, Gladys Tyen, Jason Lee, Felix Weissenberger, Trevor Strohman, Ashwin Balakrishna, Jack Rae, Marko Velic, Raoul de Liedekerke, Oded Elyada, Wentao Yuan, Canoe Liu, Lior Shani, Sergey Kishchenko, Bea Alessio, Yandong Li, Richard Song, Sam Kwei, Orion Jankowski, Aneesh Pappu, Youhei Namiki, Yenai Ma, Nilesch Tripuraneni, Colin Cherry, Marissa Ikonomidis, Yu-Cheng Ling, Colin Ji, Beka Westberg, Auriel Wright, Da Yu, David Parkinson, Swaroop Ramaswamy, Jerome Connor, Soheil Hassas Yeganeh, Snchit Grover, George Kenwright, Lubo Litchev, Chris Apps, Alex Tomala, Felix Halim, Alex Castro-Ros, Zefei Li, Anudhyan Boral, Pauline Sho, Michal Yarom, Eric Malmi, David Klinghoffer, Rebecca Lin, Alan Ansell, Pradeep Kumar S, Shubin Zhao, Siqi Zuo, Adam Santoro, Heng-Tze Cheng, Solomon Demmessie, Yuchi Liu, Nicole Brichtova, Allie Culp, Nathaniel Braun, Dan Graur, Will Ng, Nikhil Mehta, Aaron Phillips, Patrik Sundberg, Varun Godbole, Fangyu Liu, Yash Katariya, David Rim, Mojtaba Seyedhosseini, Sean Ammirati, Jonas Valfridsson, Mahan Malihi, Timothy Knight, Andeep Toor, Thomas Lampe, Abe Ittycheriah, Lewis Chiang, Chak Yeung, Alexandre Fréchette, Jinmeng Rao, Huisheng Wang, Himanshu Srivastava, Richard Zhang, Rocky Rhodes, Ariel Brand, Dean Weesner, Ilya Figotin, Felix Gimeno, Rachana Fellingner, Pierre Marcenac, José Leal, Eyal Marcus, Victor Cotruta, Rodrigo Cabrera, Sheryl Luo, Dan Garrette, Vera Axelrod, Sorin Baltateanu, David Barker, Dongkai Chen, Horia Toma, Ben Ingram, Jason Riesa, Chinmay Kulkarni, Yujing Zhang, Hongbin Liu, Chao Wang, Martin Polacek, Will Wu, Kai Hui, Adrian N Reyes, Yi Su, Megan Barnes, Ishaan Malhi, Anfal Siddiqui, Qixuan Feng, Mihai Damaschin, Daniele Pighin, Andreas Steiner, Samuel Yang, Ramya Sree Boppana, Simeon Ivanov, Arun Kandoor, Aditya Shah, Asier Mujika, Da Huang, Christopher A. Choquette-Choo, Mohak Patel, Tianhe Yu, Toni Creswell, Jerry Liu, Catarina Barros, Yasaman Razeghi, Aurko Roy, Phil Culliton, Binbin Xiong, Jiaqi Pan, Thomas Strohmann, Tolly Powell, Babi Seal, Doug DeCarlo, Pranav Shyam, Kaan Katircioglu, Xuezhi Wang, Cassidy Hardin, Immanuel Odisho, Josef Broder, Oscar Chang, Arun Nair, Artem Shtefan, Maura O'Brien, Manu Agarwal, Sahitya Potluri, Siddharth Goyal, Amit Jhinal, Saksham Thakur, Yury Stuken, James Lyon, Kristina Toutanova, Fangxiaoyu Feng, Austin Wu, Ben Horn, Alek Wang, Alex Cullum, Gabe Taubman, Disha Shrivastava, Chongyang Shi, Hamish Tomlinson, Roma Patel, Tao Tu, Ada Maksudaj Oflazer, Francesco Pongetti, Mingyao Yang, Adrien Ali Taïga, Vincent Perot, Nuo Wang Pierse, Feng Han, Yoel Drori, Iñaki Iturrate, Ayan Chakrabarti, Legg Yeung, Dave Dopson, Yi ting Chen, Apoorv Kulshreshtha, Tongfei Guo, Philip Pham, Tal Schuster, Junquan Chen, Alex Polozov, Jinwei Xing, Huanjie Zhou, Praneeth Kacham, Doron Kukliansky, Antoine Miech, Sergey Yaroshenko, Ed Chi, Sholto Douglas, Hongliang Fei, Mathieu Blondel, Preethi Myla, Lior Madmoni, Xing Wu, Daniel Keysers, Kristian Kjems, Isabela Albuquerque, Lijun Yu, Joel D'sa, Michelle Plantan, Vlad Ionescu, Jaume Sanchez Elias, Abhirut Gupta, Manish Reddy Vuyyuru, Fred Alcober, Tong Zhou, Kaiyang Ji, Florian Hartmann, Subha Puttagunta, Hugo Song, Ehsan Amid, Anca Stefanoiu, Andrew Lee, Paul Pucciarelli, Emma Wang, Amit Raul, Slav Petrov, Isaac Tian, Valentin Anklin, Nana Nti, Victor Gomes, Max Schumacher, Grace Vesom, Alex Panagopoulos, Konstantinos Bousmalis, Daniel Andor, Josh Jacob, Yuan Zhang, Bill Rosgen, Matija Kecman, Matthew Tung, Alexandra Belias, Noah Goodman, Paul Covington, Brian Wieder, Nikita Saxena, Elnaz Davoodi, Muhuan Huang, Sharath Maddineni, Vincent Roulet, Folawiyo Campbell-Ajala, Pier Giuseppe Sessa, Xintian, Wu, Guangda Lai, Paul Collins, Alex Haig, Vytenis Sakenas, Xiaowei Xu, Marissa Giustina, Laurent El Shafey, Pichi Charoenpanit, Shefali Garg, Joshua Ainslie, Boone Severson, Montse Gonzalez Arenas, Shreya Pathak, Sujee Rajayogam, Jie Feng, Michiel Bakker, Sheng Li, Nevan Wichers, Jamie Rogers, Xinyang Geng, Yeqing Li, Rolf Jagerman, Chao Jia, Nadav Olmert, David Sharon, Matthew Mauger, Sandeep Mariserla, Hongxu

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A Multilingual Synthetic Data Generation for Long Context

In this section, we provide the question-generation prompts used in the first pass of our pipeline to construct multilingual synthetic question-answer pairs, together with their corresponding reference answers. We also summarize the filtering and verification heuristics used in the second pass. Specifically, we retain only sufficiently long, self-contained documents, remove malformed text, boilerplate-heavy pages, and near-duplicate documents using fuzzy deduplication, and then verify candidate synthetic examples with an LLM-as-a-judge step using Qwen3-Next-80B-A3B-Thinking. This verification rejects malformed outputs, incomplete answers, unsupported answers not justified by the source document, weak answer-document grounding, and questions that are answerable from a short local span rather than requiring evidence aggregation across the document. These prompts define the procedures for generating summary questions, multi-part information-retrieval questions, multi-hop reasoning questions, comprehensive questions, and classification questions.

Prompt for Document Summary Question Generation

You are given the following document in {language}:

{document}

Your task is to create one **high-quality summary question** about the document and its corresponding answer.

Instructions:

- The question must require **synthesizing information from the entire document**.
- The answer must provide a **concise but complete summary** that captures all major themes, ideas, and relationships, reflecting a full and accurate understanding of the text.
- Your output must follow exactly the format below and should be in {language}.
- Do not include anything else.

Output Format:

Question: <question>

Answer: <answer>

Prompt for Multi-Part Information Retrieval Question Generation

You are given the following document in {language}:

{document}

Your task is to create one **high-quality information-retrieval question** about the document and its corresponding answer.

Instructions:

- The question must require locating and **extracting information from multiple parts** of the document.
- Ensure that it can only be answered by someone who has **thoroughly read and understood the full text**.
- The answer must provide the **precise information requested**, directly grounded in the document.
- Your output must follow exactly the format below and should be in {language}.
- Do not include anything else.

Output Format:

Question: <question>

Answer: <answer>

Prompt for Multi-Hop Reasoning Question Generation

You are given the following document in {language}:

{document}

Your task is to create one **high-quality multi-hop reasoning question** about the document and its corresponding answer.

Instructions:

- The question must require **reasoning across multiple parts** of the document.
- It should combine information from **different sections, themes, or ideas**, requiring logical synthesis of the full text.
- The answer must clearly **explain the reasoning steps** or connections involved.
- Your output must follow exactly the format below and should be in {language}.
- Do not include anything else.

Output Format:

Question: <question>

Answer: <answer>

Prompt for Comprehensive Question Generation

You are given the following document in {language}:

{document}

Your task is to create one **high-quality question** and its corresponding answer.

Instructions:

- The question must require information from the **entire document**, not just a small part.
- It should involve **multiple aspects or themes** so that it can only be answered by someone who has fully read and understood the document.
- Your output must follow exactly the format below and should be in {language}.
- Do not include anything else.

Output Format:

Question: <question>

Answer: <answer>

Prompt for Document Classification Generation

You are given the following document in {language}:

{document}

Your task is to create one **high-quality classification question** about the document and its corresponding answer.

Instructions:

- The question must require **identifying or inferring categories, labels, or sentiment** from content spread across the entire document.
- Ensure that the classification **cannot be made without reading and understanding the full text**.
- The answer must clearly provide the correct classification or sentiment label and show that it is grounded in the document’s overall content.
- Your output must be in {language} and follow the format below exactly.
- Do not include anything else.

Output Format:

Question: <question>

Answer: <answer>

B LLM-As-A-Judge Prompt for Reward Computation

During the RL stage, this prompt was given to Qwen3-4B, which acted as the LLM judge to evaluate and reward the correctness of the student model’s responses against the known ground-truth answers.

Prompt for Verifying LLM Answer Accuracy

You will receive a **question**, a **response** generated by an LLM, and a **ground-truth answer**.

Your task is to determine whether the response is **accurate** according to the ground-truth.

Instructions:

- Respond with **only a single integer**:
 - **1** → The response is accurate and aligns with the ground-truth.
 - **0** → The response is inaccurate or does not match the ground-truth.
- Do not include any explanations, text, or commentary—just the integer.

Input Format:

Question: {question}

Response: {response}

Ground-truth: {ground-truth}

Output Format:

Answer: <int>

C Detailed Results

This appendix lists the full score breakdowns for all benchmarks used in the main paper. Long-context results are reported in Section C.1, with detailed scores in Table 4 (RULER), Table 5 (LongBench), and Table 6 (∞ Bench). Multilingual short-context results are reported in Section C.2, with per-language scores in Table 7 (MMMLU) and Table 8 (MGSM). Multilingual long-context results are also reported in Section C.2, with per-language scores in Table 9 (OneRULER). In addition, Section C.4 reports the ablation on balancing short- and long-context data, Section C.5 reports the cold-start initialization ablation comparing SFT and KD, and Section C.6 reports the standard deviation across five evaluation runs for the main benchmarks and key ablations discussed in the main text.

C.1 Long-context Results

Model	Length	Avg.	S1	S2	S3	MK1	MK2	MK3	MV	MQ	VT	CWE	FWE	SQA	HQA
Qwen3 1.7B	4K	87.4	100.0	100.0	99.8	100.0	95.4	98.6	99.1	100.0	99.0	91.8	75.7	42.0	34.2
	8K	84.2	100.0	100.0	99.6	100.0	92.4	92.6	97.1	99.8	97.3	73.9	76.3	32.2	34.0
	16K	81.0	100.0	100.0	100.0	99.4	81.8	93.8	93.2	99.6	92.5	46.4	88.7	29.5	28.6
	32K	72.0	100.0	100.0	99.8	89.6	74.6	45.2	89.0	96.5	88.8	19.4	79.9	28.0	25.6
	64K	51.5	100.0	97.2	90.6	55.0	3.2	0.2	64.2	84.3	55.2	7.5	61.6	27.8	22.3
	128K	11.0	61.0	3.0	3.6	7.4	0.0	0.0	0.0	1.7	0.4	0.2	58.6	3.7	3.6
Ours	4K	88.3	100.0	100.0	99.8	99.8	97.2	99.6	91.7	100.0	99.1	90.5	82.7	47.0	40.2
	8K	85.4	100.0	100.0	99.8	99.6	93.2	97.4	94.7	99.9	99.1	83.5	73.1	34.4	36.0
	16K	84.7	100.0	100.0	97.0	99.6	90.0	97.4	92.3	100.0	91.5	77.5	88.2	32.5	35.0
	32K	80.4	100.0	100.0	98.2	99.6	91.6	80.4	91.3	98.7	80.7	71.2	67.1	32.1	33.8
	64K	65.6	100.0	100.0	98.6	85.6	32.0	22.4	92.7	95.0	70.4	48.3	44.6	35.7	27.2
	128K	44.5	76.8	66.0	65.8	45.6	13.2	2.6	44.9	49.0	73.9	27.6	78.1	19.6	15.4

Table 4: RULER (Hsieh et al., 2024) breakdown across sequence lengths.

Category	Qwen3-1.7B	Ours
<i>Single-Document QA</i>	34.3	39.2
<i>Multi-Document QA</i>	23.4	43.1
<i>Summarization</i>	20.0	20.2
<i>Few-shot Learning</i>	42.9	43.3
<i>Synthetic Tasks</i>	65.3	65.0
<i>Code Completion</i>	20.3	26.9
Average	34.4	39.6

Table 5: LongBench (Bai et al., 2024b) results by task category.

Category	Qwen3-1.7B	Ours
<i>R.PassKey</i>	13.9	100.0
<i>R.Number</i>	15.4	93.2
<i>Retrieve.KV</i>	0.0	0.0
<i>En.Sum</i>	11.5	11.8
<i>En.QA</i>	0.1	8.1
<i>Zh.QA</i>	3.9	5.7
<i>En.MC</i>	18.8	30.1
<i>En.Dia</i>	0.5	1.2
<i>Code.Debug</i>	3.7	4.6
<i>Code.Run</i>	0.5	1.5
<i>Math.Calc</i>	6.0	6.0
<i>Math.Find</i>	4.9	10.6
Average	6.6	22.7

Table 6: ∞ Bench (Zhang et al., 2024) results by task, including all benchmark categories. Many low scores (e.g., *Retrieve.KV*, *En.QA*, *Code.Run*, *Math.Calc*) reflect the difficulty of reasoning over extremely long contexts ($>100K$ tokens), which current models, especially smaller LLMs (e.g., 1.7B parameters), still struggle with.

C.2 Multilingual Results

Language	Qwen3-1.7B	Ours
<i>ar</i>	53.2	54.1
<i>ca</i>	60.6	60.5
<i>da</i>	58.8	58.0
<i>de</i>	61.1	62.8
<i>es</i>	62.7	62.8
<i>fr</i>	63.0	62.4
<i>hi</i>	49.7	50.6
<i>hu</i>	56.2	56.2
<i>it</i>	62.7	62.9
<i>nl</i>	61.3	61.0
<i>pt</i>	62.7	63.5
<i>ro</i>	61.2	60.5
<i>ru</i>	61.4	61.6
<i>sv</i>	60.5	60.3
<i>uk</i>	58.5	60.2
<i>zh</i>	55.4	55.8
Average	59.3	59.6

Table 7: MMMLU (OpenAI, 2024) results by language.

Language	Qwen3-1.7B	Ours
<i>de</i>	80.8	79.2
<i>es</i>	79.2	80.0
<i>fr</i>	81.2	80.8
<i>ja</i>	52.8	57.6
<i>ru</i>	70.8	70.8
<i>zh</i>	70.4	71.2
Average	72.5	73.3

Table 8: MGSM (Shi et al., 2022) results by language.

Language	Qwen3-1.7B					Ours				
	Avg.	8K	32K	64K	128K	Avg.	8K	32K	64K	128K
<i>de</i>	63.3	99.0	84.8	69.5	0.1	75.1	99.0	97.5	74.1	29.6
<i>es</i>	63.8	98.4	85.0	70.8	1.0	78.3	99.3	96.6	74.1	43.3
<i>fr</i>	63.4	98.9	85.4	68.8	0.5	76.8	99.0	96.5	71.6	40.1
<i>it</i>	62.8	98.9	84.5	67.0	0.8	75.3	99.0	97.1	71.1	34.1
<i>ja</i>	64.9	99.0	86.4	74.2	0.0	73.9	99.0	96.3	79.4	21.0
<i>ko</i>	83.8	98.9	86.4	74.9	7.9	86.5	98.8	96.4	77.9	72.9
<i>nl</i>	62.4	98.1	83.4	68.1	0.1	74.9	98.8	97.6	71.5	31.5
<i>pt</i>	63.9	98.4	85.1	71.6	0.5	77.8	98.9	97.1	76.0	39.1
<i>ru</i>	63.7	99.0	85.4	68.9	1.5	78.3	98.9	96.3	74.0	43.8
<i>zh</i>	64.7	98.5	86.4	74.0	0.0	74.1	98.6	94.8	78.8	24.0
Average	65.7	98.7	85.3	70.8	1.2	77.1	98.9	96.6	74.9	37.9

Table 9: OneRULER (Kim et al., 2025) results by language.

C.3 Language-wise Composition of LONGBLOCKS

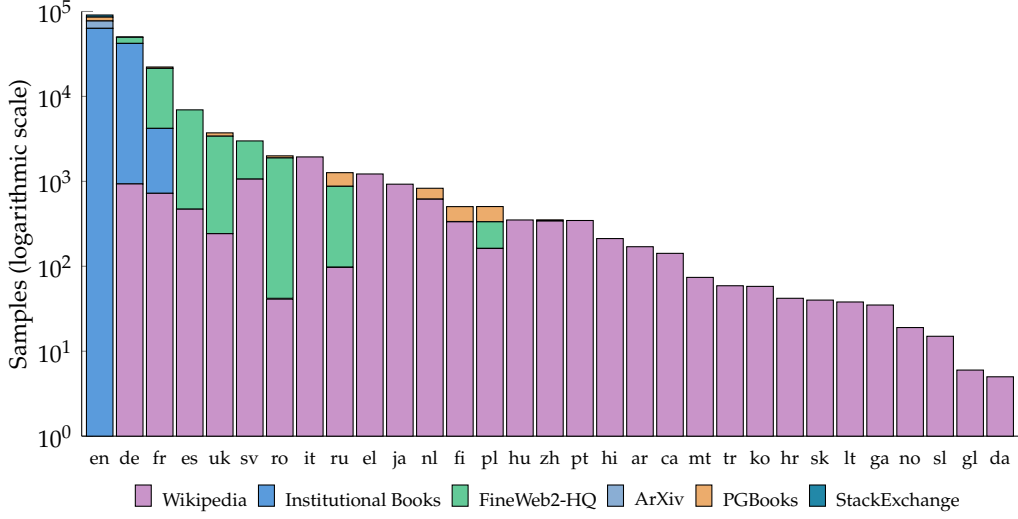


Figure 7: Language-wise composition of the corpora used to generate multilingual synthetic long-context data for LONGBLOCKS. Each stacked bar corresponds to one language, and colored segments denote the number of samples contributed by each source. Programming and markup languages are omitted for readability. These languages are sampled uniformly from Stack-Edu.

C.4 Balancing Short- and Long-Context Data

We ablate the mix of short-context data and LONGBLOCKS under a fixed token budget. Figure 8 shows that short-context performance is largely insensitive to the ratio, with 10% short-context data sufficient to preserve short-task accuracy. In contrast, long-context accuracy declines steadily as the share of short-context data increases. Overall, allocating 90% of training tokens to LONGBLOCKS yields the best trade-off, preserving short-context capabilities while improving long-context performance.

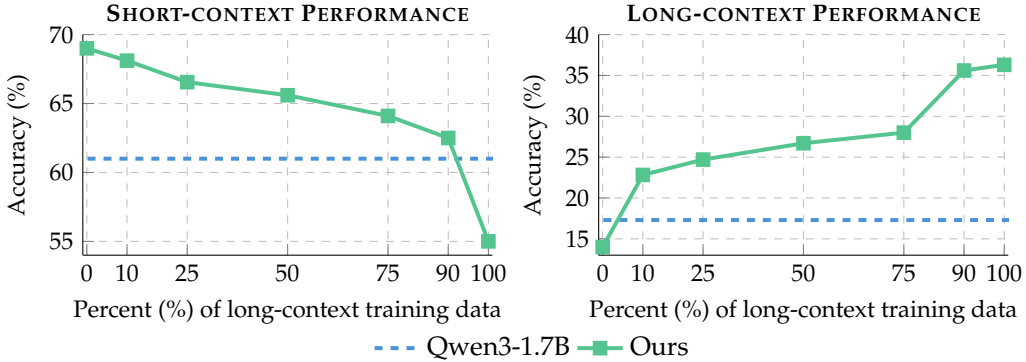


Figure 8: Accuracy on short- and long-context tasks across different short/long data mixes.

C.5 Cold-start Initialization: SFT vs. KD

We ablate the cold-start strategy for off-policy initialization, comparing SFT and KD. As shown in Figure 9, KD provides only slight gains on both short- and long-context benchmarks. Given these limited benefits and its higher cost when teacher logits are not cached, we adopt SFT as the default cold-start method.

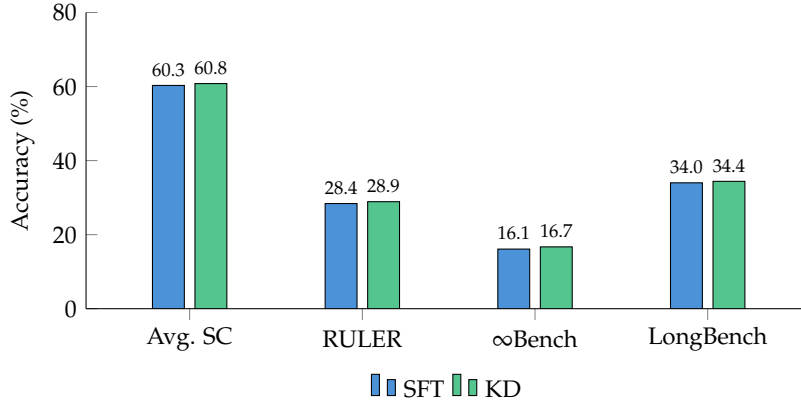


Figure 9: Comparison of SFT and KD on short-context (SC) vs. long-context benchmarks.

C.6 Uncertainty Results

Evaluation with reasoning models can be computationally expensive and subject to non-trivial run-to-run variability under stochastic decoding. To account for this variability, we evaluate all main benchmark results over five runs using the shared decoding configuration described in Section 4.1. The main text reports the mean score across runs for readability, while this section reports the standard deviation across the five runs for each benchmark. As shown in Table 10, these results indicate that our post-training approach does not introduce additional variability, with standard deviations that remain broadly in line with those of reasoning models, including Qwen3-1.7B.

Benchmark	Qwen3-1.7B	Ours
IFEval	0.6608	0.7155
IFBench	0.9901	0.6487
MMLU-Pro	0.1441	0.1427
GPQA $\blacklozenge$	1.9992	1.9499
GSM8K	0.4654	0.4461
MATH-500	0.4775	0.3578
HumanEval	0.8627	0.7550
HumanEval+	0.5561	0.4650
RULER	0.3030	0.1500
LongBench	0.2550	0.2168
∞ Bench	0.3265	0.3918

Table 10: Standard deviation across five evaluation runs for the main benchmark suite.